

Assignment: .NET Application Development

Objective:

Design and implement a Minimum Viable Product (MVP) for a House Broker Application using the .NET, MSSQL for the database, and adhering to Clean Architecture principles.

Requirements:

User Authentication:

- Implement user authentication using .NET Identity or a similar authentication mechanism.
- Differentiate between house seekers and brokers in the authentication process.

Listing Management:

- Implement CRUD operations for property listings, ensuring data integrity in the MSSQL database.
- Include essential details such as property type, location, price, and features.
- Expose the API so can be used by third party
- Add proper validation.
- Implement caching.

Implement a **commission engine** for Brokers:

Each property listing should calculate an **estimated broker commission** using the following logic:

- If price < 50,00,000 → 2%
- If price between 50,00,000 and 1 crore → 1.75%
- If price > 1 crore → 1.5%
- The commission amount should be stored/displayed with the listing (but only visible to the broker who owns the listing).
- Make commission rates configurable via the DB.

Show Case:

- Implement a search functionality for users to find properties based on criteria such as location, price range, and property type.

- Provide filters to refine search results.
- Retrieve and display detailed information about each property from the MSSQL database, including images, description, and contact details of the broker.

Unit Testing:

- Write Unit tests for
 - Commission logic
 - Listing service logic
 - Basic controller tests with mocked dependencies

Submission Guidelines:

- Provide well-documented source code following Clean Architecture principles with clear comments explaining the functionality of each layer.
- Include a README file with instructions for setting up and running the application, including any necessary database setup.

Note: Please don't hesitate to seek clarification on any requirements. Good luck!